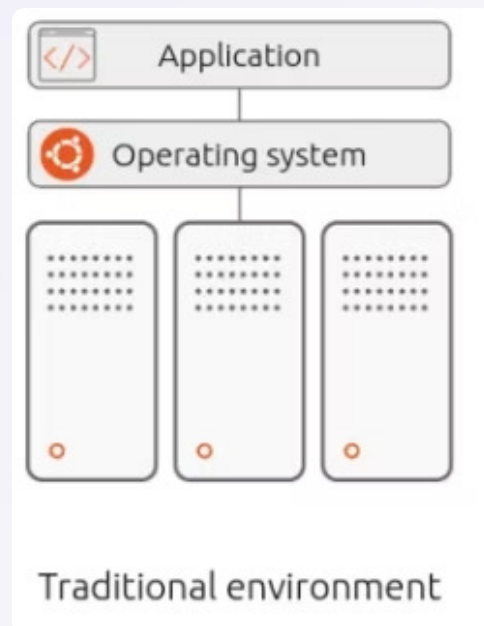


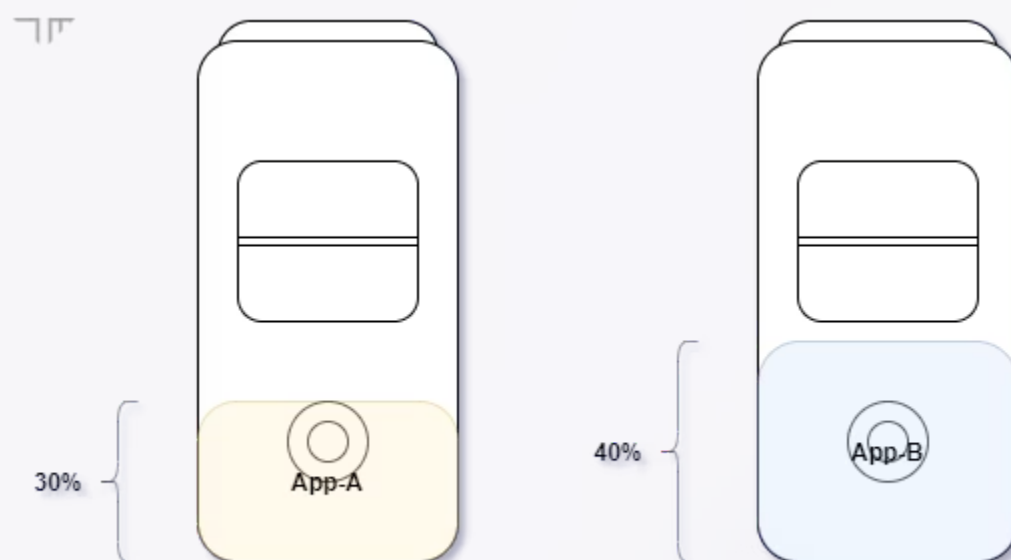
VIRTUALIZATION!!!



Suppose company 'XYZ' needs to run 2 different applications — App A & App B. Each app needs a particular operating system or has a set of dependencies / libraries which are not compatible with the other app.

Not a very uncommon scenario.

In a world without virtualization, the company naturally gets 2 different physical machines for each app.



Running App A and B on two different machines

Notice that, App A utilizes 30% of the first machine while App B utilizes 40% of the second.

We can immediately note the major downsides to this approach:

- they need to buy additional hardware for each new application
- each app is not fully utilizing the physical resources available i.e. waste of resources
- the number of physical resources needed to manage grows linearly with the number of applications needed
- security needs to be maintained in every single system individually

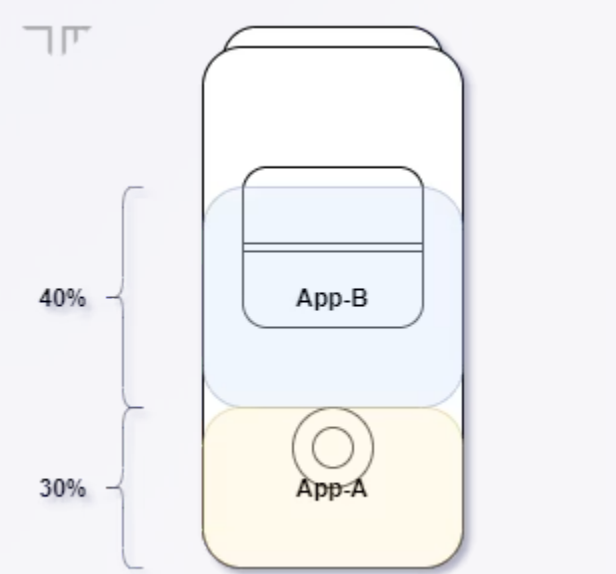
In absence of a solution, the downsides and higher overhead might not be trivial. Therefore, this has been the traditional way of running applications for many years.

Virtualization

Virtualization is the technology that allows us to create & manage virtual resources that are isolated from the underlying hardware.

Using virtualization, we can make a single computer act like multiple virtual ones.

Virtualization allows company XYZ to run both of their apps A & B on a single machine— saving them lots of money and pain in the 🍊.



Running App A and B on a single machine using virtualization

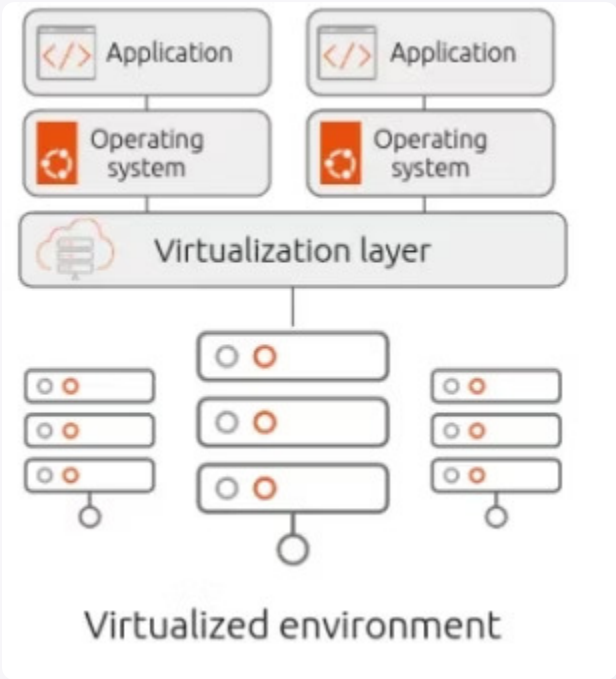
With virtualization, the single physical machine can be partitioned into several smaller ones; each of which acts as if it's a completely separate machine — therefore the term “virtual”.

A program running inside a virtual environment only has access to the resources assigned to that virtual machine or partition.

So, company XYZ could even have a third application on the same machine if it consumed less then 30% of the machines resources! (although it is recommended to not run machines at 100% load for long periods)

Virtualization

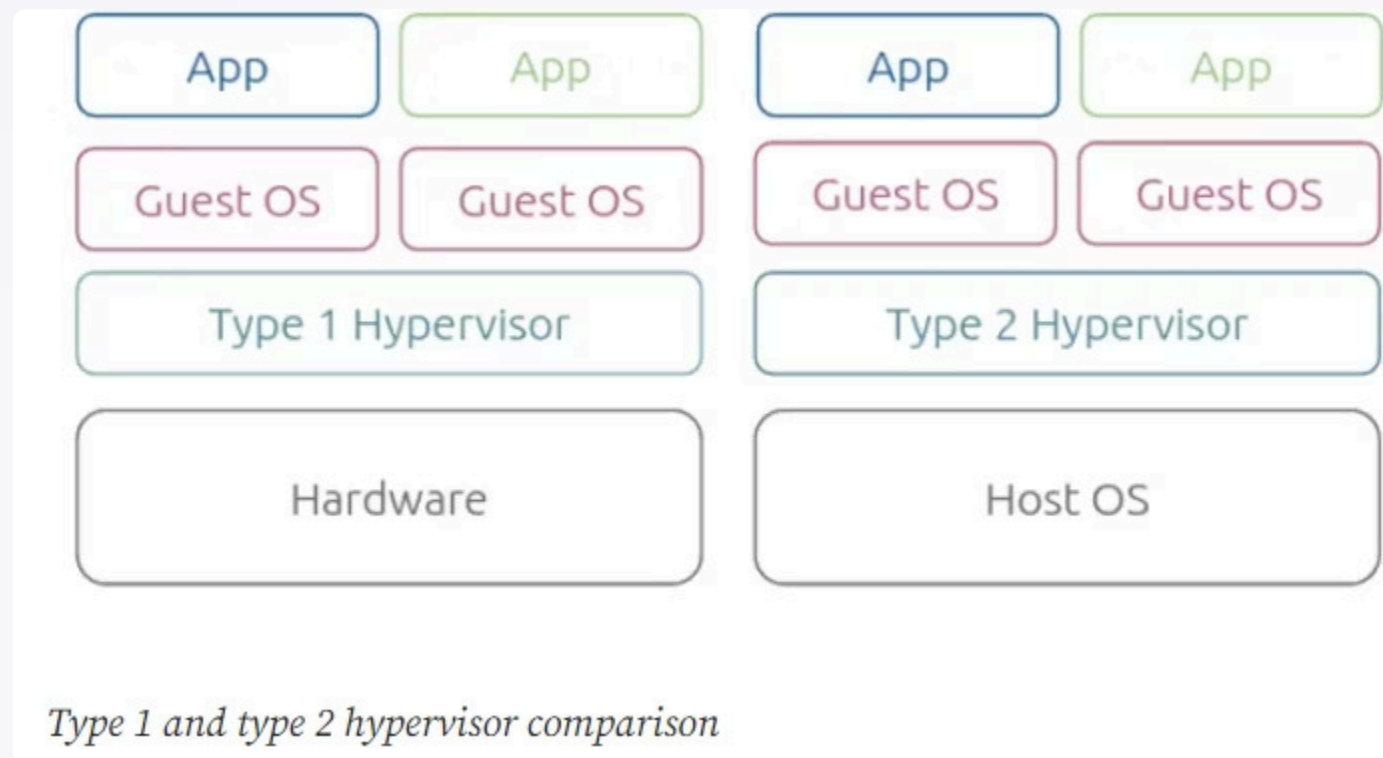
But Virtualization means rather than having the operating system run directly on top of the physical hardware, an additional virtualization layer is added in between, enabling users to deploy multiple virtual servers, each with their own operating system, on one physical machine.



The role of a hypervisor

A hypervisor (also known as a virtual machine monitor) — a software layer enabling multiple operating systems to co-exist while sharing the resources of a single hardware host.

The hypervisor acts as an intermediary between virtual machines and the underlying hardware, allocating host resources such as memory, CPU, and storage.



- Type 1 hypervisors, also known as bare-metal hypervisors, run directly on the host's hardware and are responsible for managing the hardware resources and running the virtual machines. Because they run directly on the hardware, they are often more efficient and have a smaller overhead than Type 2 hypervisors. Examples of Type 1 hypervisors include VMware ESXi, Microsoft Hyper-V, and Citrix XenServer.
- Type 2 hypervisors, also known as hosted hypervisors, run on top of a host operating system and rely on it to provide the necessary hardware resources and support. Because they run on top of an operating system, they are often easier to install and use than Type 1 hypervisors, but they might be less efficient. Examples of Type 2 hypervisors include VMware Workstation and Oracle VirtualBox.

Benefits and disadvantages

The biggest benefit of virtualization

- Cost savings.
- Resource utilization.
- Being able to run multiple VMs on a single server prevents some of your servers from standing idle.
- Easily create, destroy and migrate VMs between different hosts, making it easier to scale and manage your computing resources, as well as implement disaster recovery plans.

Disadvantages,

- Virtualization does increase the performance overhead, given that it introduces an additional layer between the host and the operating system.
- Depending on the workload, the decrease in performance can be noticeable, unless significant RAM and CPU resources are allocated.
- In addition, while there is cost saving in the long run, the upfront investment can be burdensome.
- Virtualization also adds some level of complexity to running your infrastructure, as you do need to manage and maintain both physical and virtual instances.